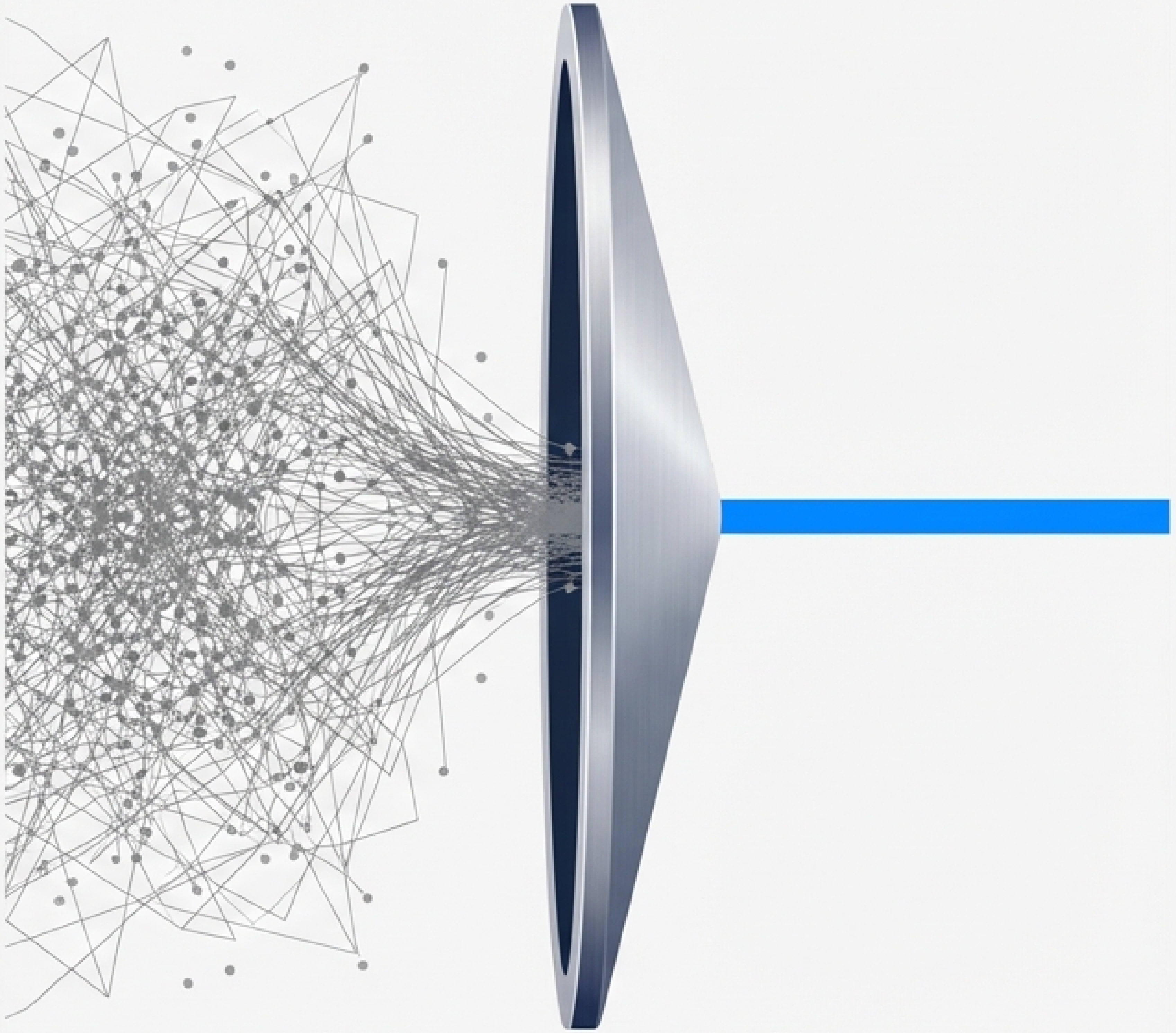
Lasso Regression: The Antidote to High-Dimensional Overfitting

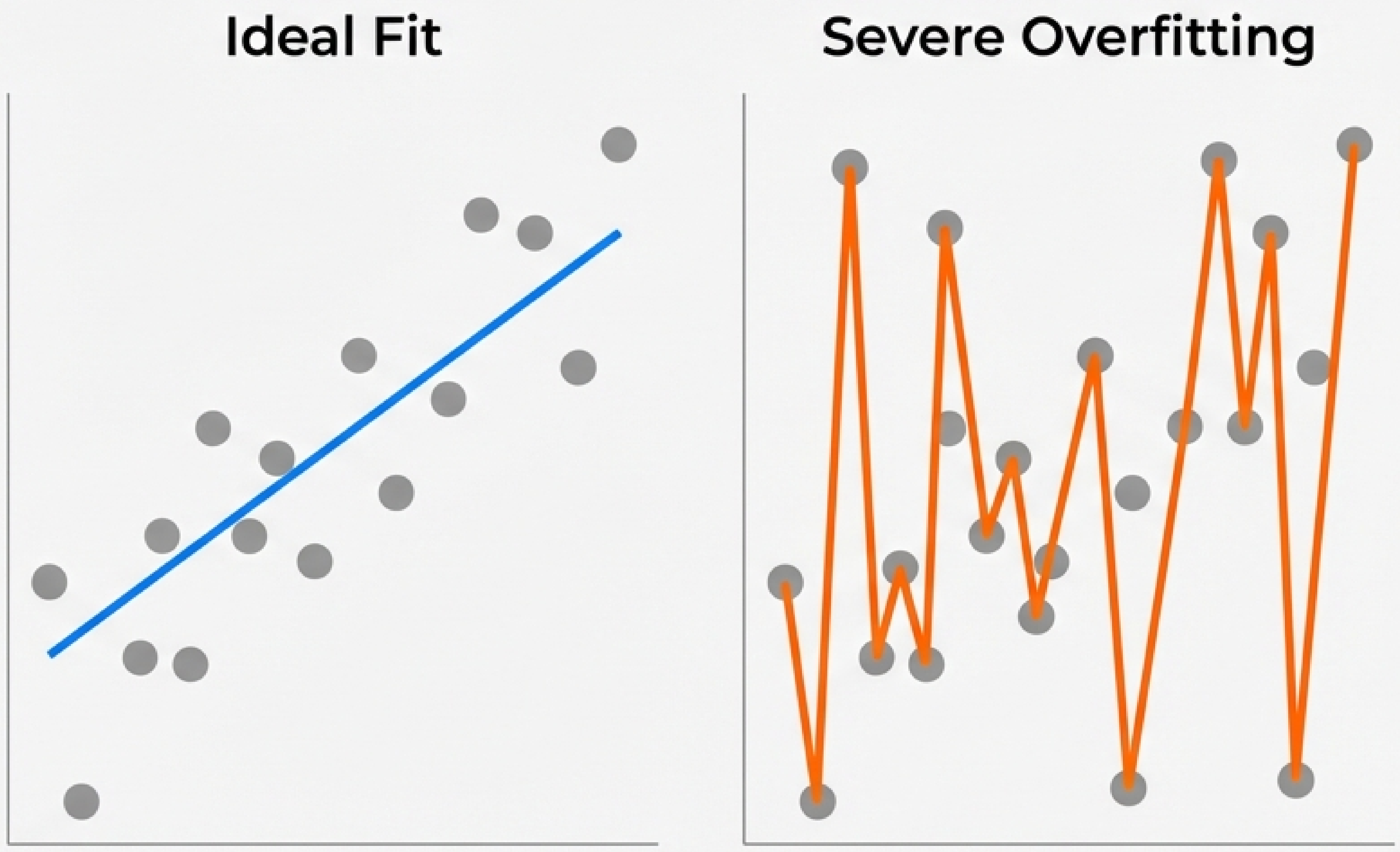
Understanding L1 Regularisation and Automatic Feature Selection



The Problem: Too Many Features, Too Much Noise

In high-dimensional data, standard models try to learn from every column.

- More columns = higher risk of the model memorising noise.
- Results in extremely high coefficients.
- Creates a fragile model that fails on new data.



Enter Lasso Regression (L1 Regularisation)

Penalises Size

Lasso adds a strict penalty to the model for being too complex.

Core Goal: To forcefully reduce overfiting by shrinking the coefficients of less important features.

Protects Against Noise

Eliminates Variables

The Engine: How It Works

$$\text{Loss} = \underbrace{\text{MSE}}_{} + \lambda \underbrace{\sum |w|}_{}$$

Mean Squared Error
The standard goal: minimise the difference between predictions and reality.

Lambda (or Alpha)
The penalty dial. Controls how strictly we punish complexity.

The L1 Norm
The absolute value of the coefficients. This is the exact penalty applied.

The Only Mathematical Difference

$$|w|$$

Lasso (L1): Uses the absolute value of the coefficients.

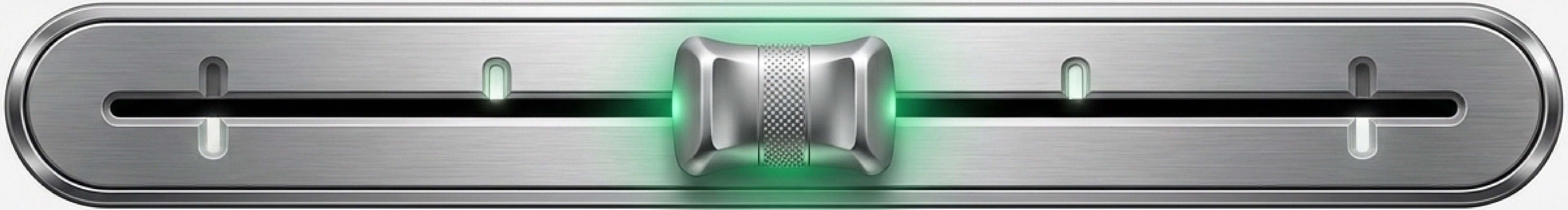
$$w^2$$

Ridge (L2): Uses the squared value of the coefficients.

This single change in the formula completely alters how the model handles data.

The Dial of Truth: Tuning Lambda (λ)

λ = 0 | **Optimal λ** | **High λ**



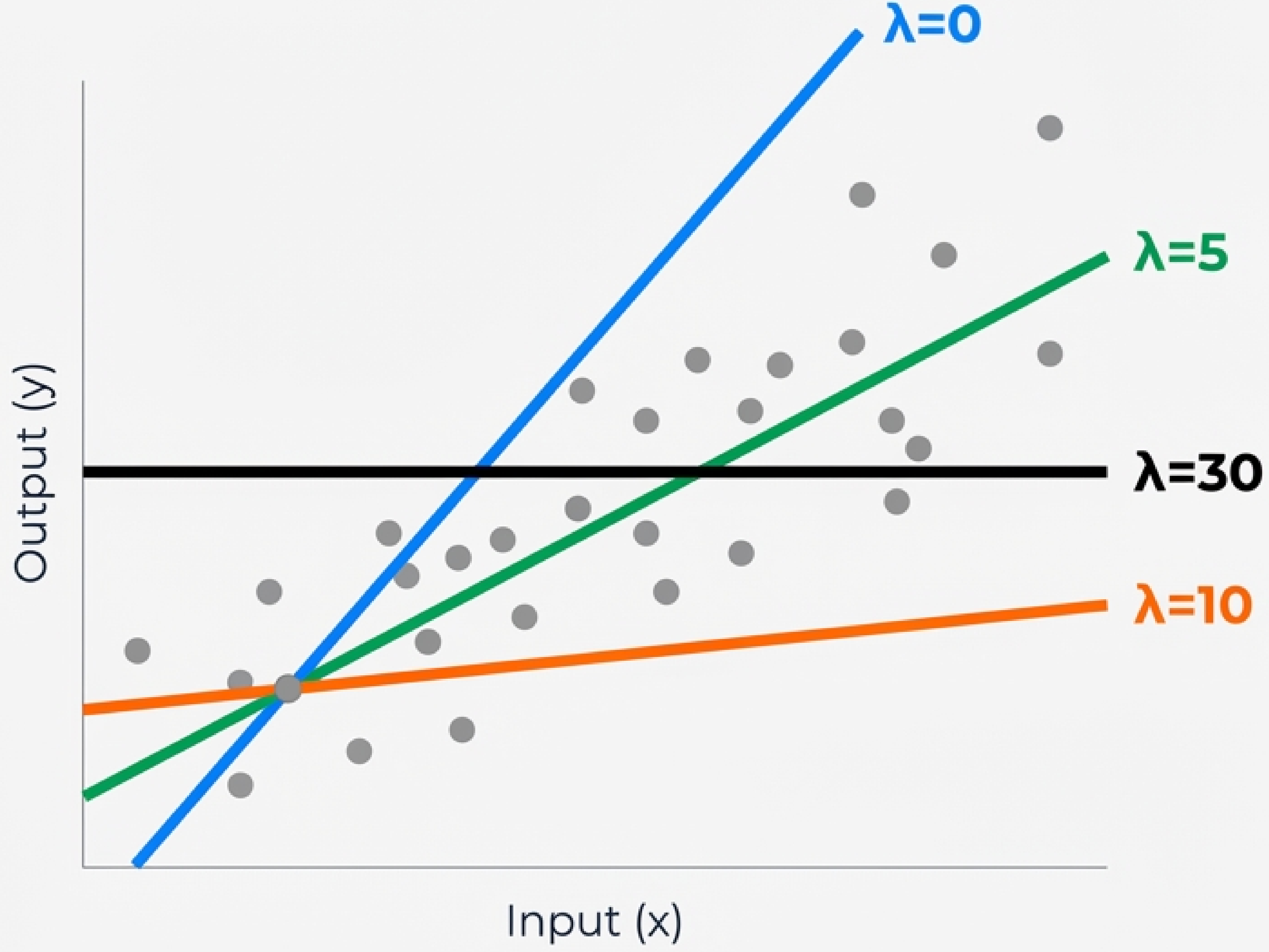
Standard Linear Regression. Zero penalty. High risk of overfitting.

The Sweet Spot. Balanced penalty.

Severe underfitting. Penalty is too high; model ignores the data entirely.

Visualising the Penalty

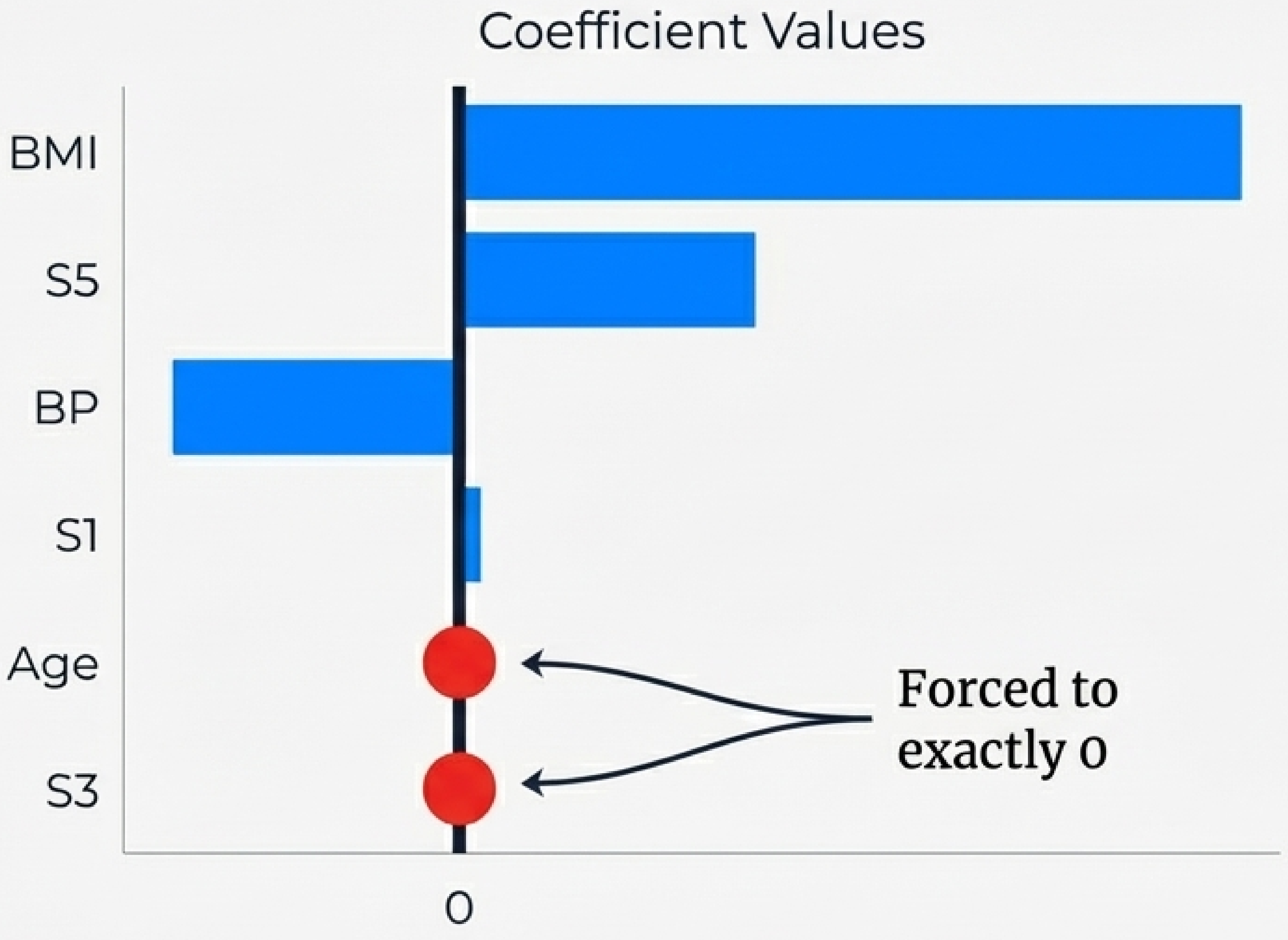
As λ increases, the model relies less on the input features (x). The slope of the line steadily decreases until it hits exactly zero (a horizontal line).



The Magic Trick: Hitting Absolute Zero

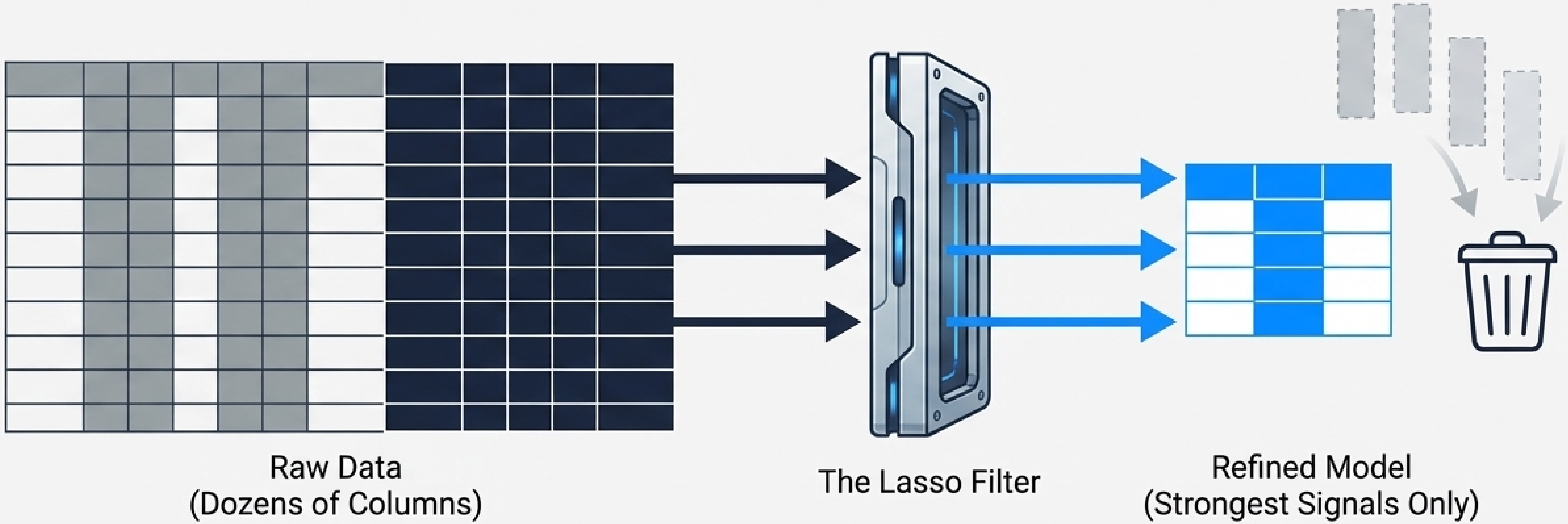
Ridge Regularisation only shrinks coefficients near zero. Lasso pushes them to exactly zero.

Data Note: In our diabetes dataset test, 'Age' and 'S3' drop entirely out of the equation.



The Superpower: Automatic Feature Selection

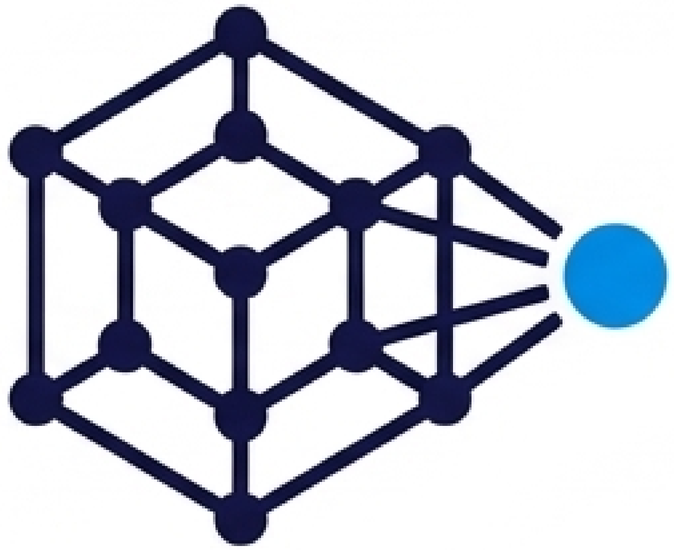
By forcing coefficients to exactly zero, Lasso tells you precisely which columns are useless. It doesn't just reduce the impact of bad features; it deletes them from the model entirely.



The High-Dimensional Advantage

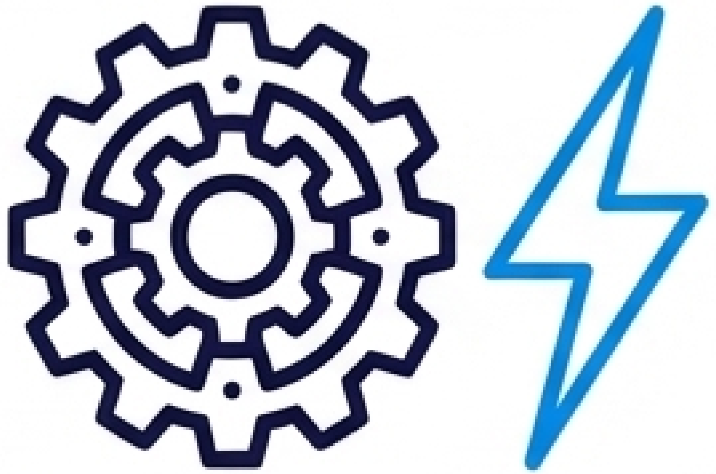
Filters Noise

Ideal for datasets with hundreds of features where only a few truly matter.



Reduces Sparsity

Decreases the dimensionality of the data automatically.

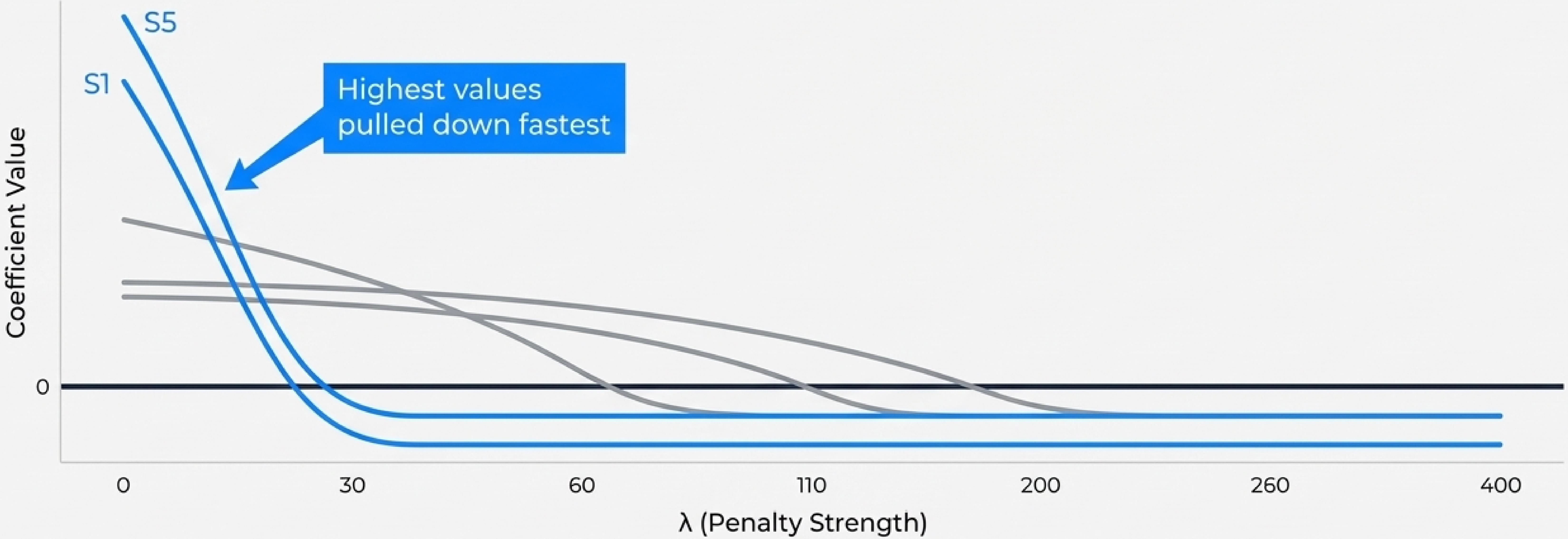


Simplifies Models

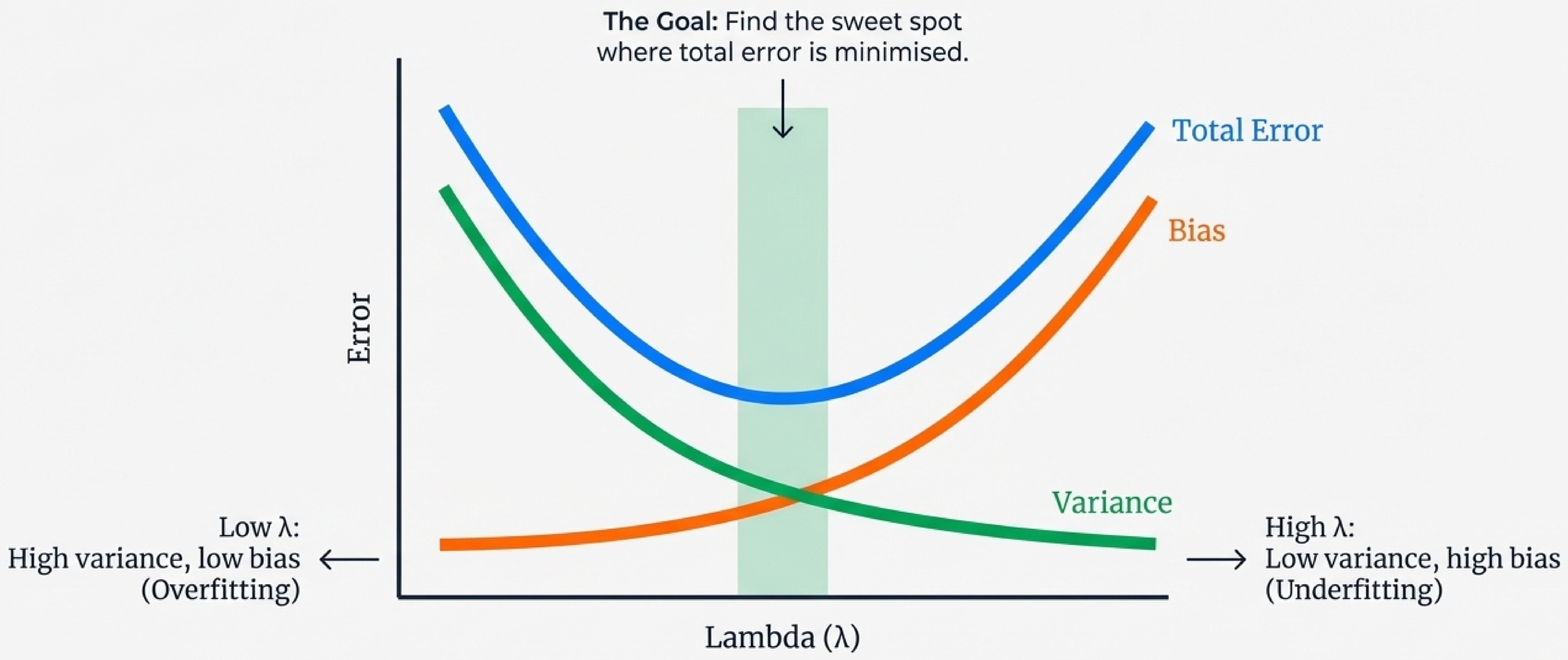
You are left with a leaner, faster, and far more interpretable model.

Higher Values, Harder Penalties

Large coefficients are penalised aggressively. Lasso forces the most dominant features to scale down down rapidly, ensuring no single variable entirely dictates the model's predictions before feature selection begins.

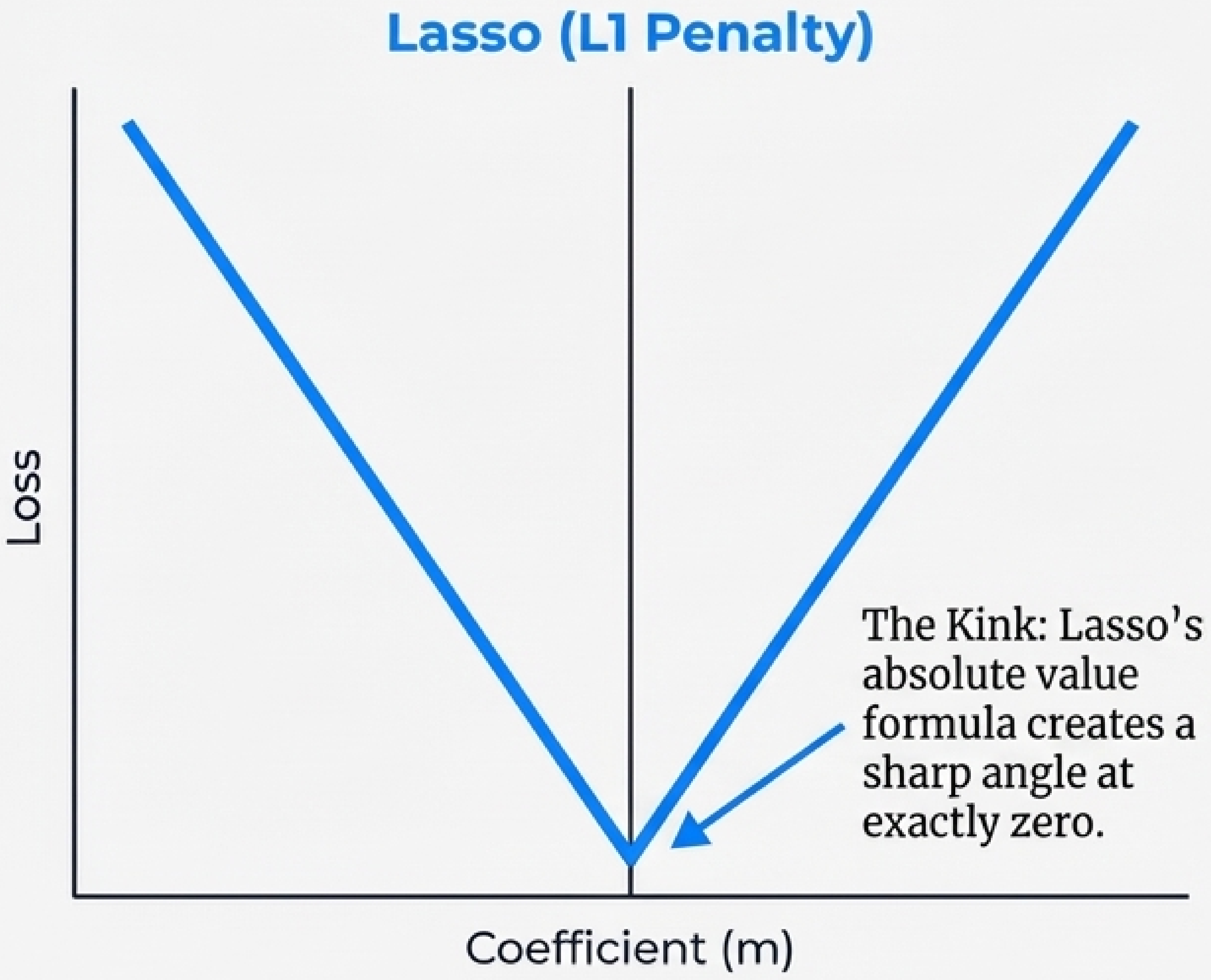


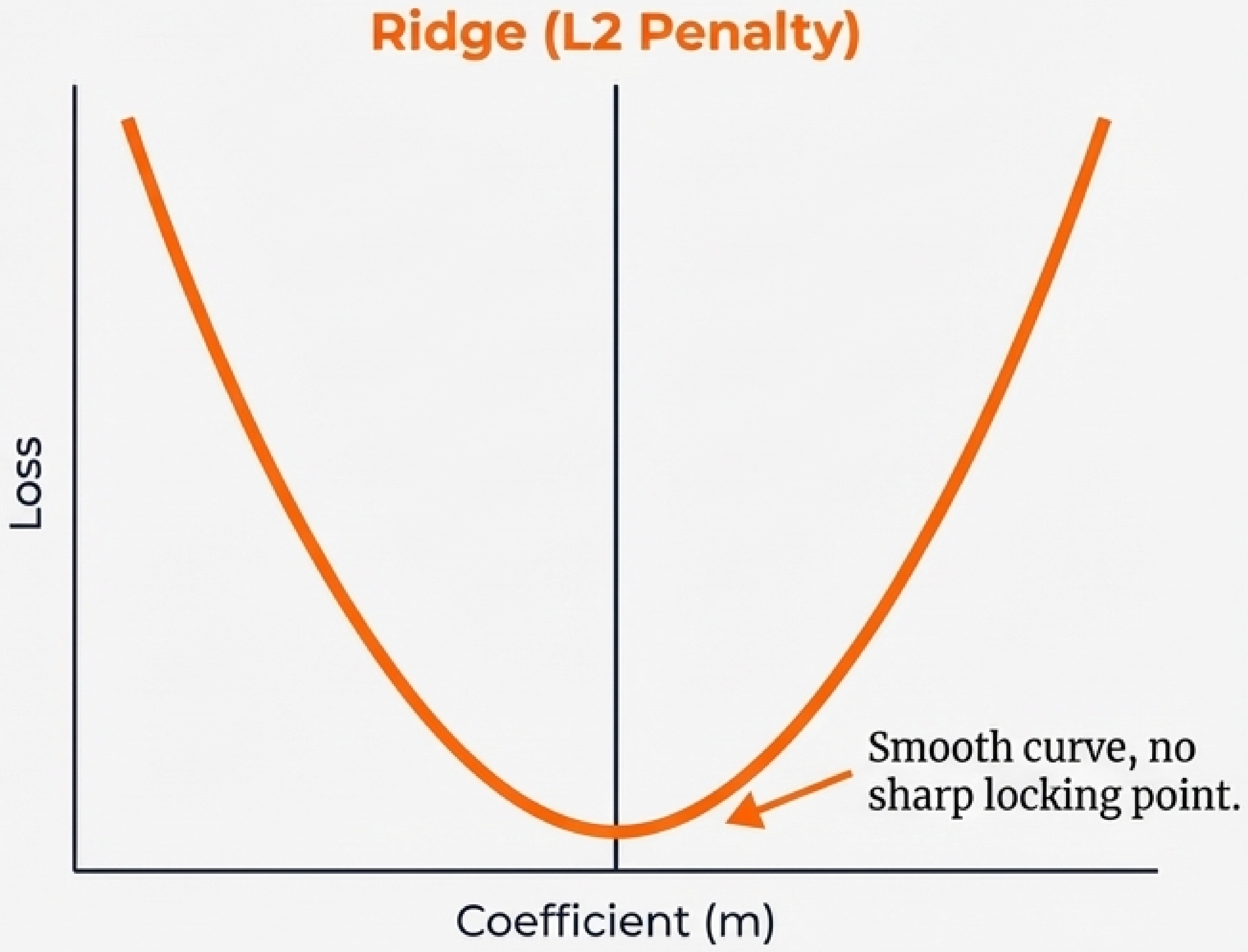
The Bias-Variance Trade-off



The Geometry of Lasso

The Result: Once the minimum loss hits zero, it locks. Increasing λ further simply flattens the curve outward from that point, keeping the coefficient permanently at zero.





The Showdown: Lasso vs. Ridge

Lasso (L1)	Ridge (L2)
Feature Selection: Yes (Automatic)	Feature Selection: No
Coefficients: Shrinks to exactly zero	Coefficients: Shrinks near zero
Best Use Case: High-dimensional data with lots of useless noise.	Best Use Case: Datasets where most features are useful or highly correlated.

Summary & Key Takeaways

- Lasso Regression (L1) reduces overfitting by penalising model complexity using absolute values.
- Its superpower is automatic feature selection—it actively zeros out the coefficients of irrelevant data columns.
- Tuning the λ (Lambda) parameter is critical to finding the sweet spot between an overfitted, noisy model and an underfitted, flatline model.